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CHEMICAL ENGINEERING DEPARTMENT

BSC. CHEMICAL ENGINEERING

DEVELOPMENT OF A SUGARCANE BAGASSE PASTEURIZATION MODEL

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ABSTRACT

Sugarcane bagasse is a widely available agricultural residue with growing potential in renewable energy and bio-based material production. However, its high moisture content and susceptibility to microbial degradation require controlled thermal treatment before value-added utilization. In many small- and medium-scale industrial applications, pasteurization is conducted using empirical heating durations without integration of thermophysical characterization or heat transfer modeling, resulting in extended processing times and limited process control.

This study developed a semi-batch pasteurization model for sugarcane bagasse to support the design of an improved pasteurization system. Thermophysical properties relevant to wet processing conditions were incorporated into transient energy balance equations, and steam behavior under direct contact conditions was considered to evaluate heat transfer performance. Simulation and sensitivity analyses were conducted to assess the influence of steam mass flowrate on temperature rise and pasteurization time.

The findings demonstrate the importance of property-based modeling in predicting pasteurization duration and improving system efficiency. The developed framework provides a foundation for transitioning from empirical operation to model-based design in biomass pasteurization systems for industrial bio-composite production.

TABLE OF CONTENTS

List of Abbreviations

SCB – Sugarcane Bagasse.

DWSIM – Simulation Software.

FAO – Food and Agriculture Organization.

MDF – Medium Density Fireboard.

KNBS – Kenya National Bureau of Statistics.

EU – European Union.

List of Charts

List of Figures

List of Equations

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

Globally, the rise in agricultural productivity and agro-industrial processing has led to a significant increase in biomass residues. These by-products, including sugarcane bagasse, rice husks, and coconut shells, present both a disposal challenge and an opportunity for sustainable development. In recent years, global efforts have focused on transforming these residues into useful products to support circular economy practices, mitigate waste, and generate additional income for farmers and processing industries.

One of the most abundant biomass residues is sugarcane bagasse, the fibrous material that remains after extracting juice from sugarcane stalks. In Kenya, sugarcane farming is a central economic activity, particularly in Western and Nyanza regions, where favorable climatic conditions and extensive land availability support large-scale cultivation. As the demand for sugar continues to grow locally and regionally, the volume of bagasse generated has increased significantly. According to Verte Solutions (2021), bagasse constitutes a substantial portion of Kenya's agro-industrial waste and is often underutilized. When improperly managed, accumulated untreated bagasse becomes a breeding ground for mold, bacteria, and pests, leading to unpleasant odors and degradation that limits its reuse (Verte Solutions, 2021).

Industrially, sugar factories in Kenya have traditionally used bagasse as a biomass fuel for internal energy generation. Through cogeneration systems, these factories produce steam and electricity, contributing up to 3 MW of power in some facilities (Barasa Kabeyi, 2023). Beyond energy generation, bagasse serves as a cost-effective alternative to wood pulp in paper production and is incorporated into compost, animal feed, and biodegradable packaging materials (Hossam & Fahim, 2023). At smaller scales, innovative enterprises have demonstrated the suitability of treated bagasse for producing eco-friendly building panels and insulation materials for low-cost housing projects in urban and peri-urban areas (Verte Solutions, 2021).

Despite its versatility, bagasse is highly perishable due to its high moisture content and rich organic composition. If not promptly treated, it undergoes microbial decomposition within days, making it unsuitable for further use. Traditional disposal practices such as open-air dumping and burning pose serious environmental and health risks, including air pollution, greenhouse gas emissions, and respiratory hazards to surrounding communities (Yakazi Network, 2024). The environmental costs associated with these practices highlight the urgent need for controlled and sustainable treatment methods.

Pasteurization is a critical pre-treatment step in preparing bagasse for controlled reuse in manufacturing processes. It reduces harmful microorganisms and stabilizes the material before further processing. However, in many small-scale operations, pasteurization systems rely on empirical heating durations without proper temperature monitoring, steam control, or predictive modeling. Such systems may lead to long processing times, inefficient steam utilization, and inconsistent thermal performance.

Although pasteurization is essential for stabilizing bagasse prior to reuse, existing industrial system largely rely on empirical heating durations without incorporating heat transfer analysis or thermos-physical property data. This results in inefficient steam utilization, prolonged processing time, and inconsistent thermal performance. Therefore, there is a need to develop a scientifically grounded pasteurization model that integrates thermos-physical properties of bagasse, heat transfer principles, and steam thermodynamics to optimize system design and performance.

1.2 Problem Statement

Despite the growing need for stabilized and reusable bagasse in agro-industrial applications, the current industrial pasteurization system operates using a fixed-time approach rather than a model-based method grounded in heat transfer principles. Steam supply is not optimized, and key thermos-physical properties such as thermal conductivity, specific heat capacity, moisture content, and bulk density are not incorporated into system analysis. As a result, pasteurization requires extended processing periods, steam consumption may be inefficient, and production capacity remains limited. A predictive semi-batch pasteurization model is therefore required to estimate temperature profiles, determine optimal steam flow rate, and inform the design of an improved pasteurizer system.

1.3 Aim of the Study

This study aimed to develop a semi-batch pasteurization model for sugarcane bagasse and use the model to propose an improved pasteurizer system design.

1.4 Objectives of the Study

The objectives of this study were:

To determine the thermal conductivity, specific heat capacity, moisture content, and bulk density of sugarcane bagasse.

To develop a semi-batch heat transfer model for steam pasteurization.

To incorporate steam thermodynamics and condensation behavior into the model.

To perform a sensitivity analysis on the steam flow rate.

To propose a pasteurizer system design based on model results.

1.5 Justification of the Study

Improving bagasse treatment aligns with sustainable waste management and value addition strategies highlighted in recent studies (Hossam & Fahim, 2023; Verte Solutions, 2021). A model-based pasteurization system reduces reliance on empirical heating times and allows prediction of pasteurization duration based on thermodynamic and heat transfer principles. Optimizing steam utilization also supports improved energy efficiency, complementing existing biomass-based energy practices in Kenya's sugar industry (Barasa Kabeyi, 2023).

By transitioning from a fixed-time empirical system to a scientifically designed semi-batch process, this study contributes to enhanced process control, reduced environmental risks associated with improper disposal (Yakazi Network, 2024), and scalable biomass utilization.

1.6 Scope of the Study

This study focused on wet, uncompacted sugarcane bagasse subjected to semi-batch steam pasteurization. The lumped capacitance assumption was applied following the Biot number evaluation. The steam flow rate was varied within a defined operational range to assess system sensitivity.

The study was limited to model development, simulation, and system design proposal. Detailed economic analysis, computational fluid dynamics modeling, and full industrial-scale implementation were beyond the scope of this work.

CHAPTER TWO: LITERATURE REVIEW

2.0 Overview

This chapter reviews the generation and utilization of sugarcane bagasse, highlighting its global and Kenyan production statistics and growing market value in energy and bio-based materials. It then examines current industrial treatment practices, particularly pasteurization, and outlines the thermal and operational challenges associated with processing wet lignocellulosic biomass.

The chapter concludes by identifying gaps in thermophysical characterization and model-based pasteurization design, which form the basis for the experimental and modeling approach adopted in this study.

2.1 Sugarcane Bagasse Generation and Utilization

The global push for sustainable development and responsible waste management has intensified attention on agricultural residues, which are often overlooked despite their volume and resource potential. Agricultural systems generate substantial quantities of lignocellulosic biomass that pose environmental risks when poorly managed, but offer economic value under circular economy models (Koul et al., 2022). Among these residues, sugarcane bagasse, the fibrous by-product of sugar extraction, represents one of the most significant and underutilized biomass resources globally (FAO, 2017). Understanding its generation patterns, utilization pathways, and economic potential provides a foundation for evaluating its technical and management challenges, particularly in developing contexts (Schwerdtner Máñez et al., 2024).

Globally, sugarcane is one of the most cultivated crops, with production exceeding 1.9 billion tons in 2022 (FAO, 2023). Approximately 0.5 tons of bagasse are produced per ton of cane processed (Tahir et al., 2021), resulting in an estimated global bagasse generation exceeding 500 million tons annually (Dale, 2003). Chemically, sugarcane bagasse contains approximately 35–55% cellulose, 20–25% hemicellulose, and 18–24% lignin (Moghaddam et al., 2014; Tahir et al., 2021). This composition supports its suitability for energy production, pulp and paper manufacturing, biodegradable packaging, soil conditioning, and engineered bio-composites.

Kenya mirrors this global pattern at a smaller but economically significant scale. The national sugar industry processes between 6 and 8.7 million tonnes of cane annually, generating approximately 1.8–2.6 million tonnes of bagasse across more than fourteen mills (Verte Solutions, 2021). The sector directly involves approximately 300,000 farmers and indirectly supports over 8 million people (Dr. Muchelule Yusuf, 2024; Ministry of Agriculture, 2023), underscoring its socio-economic importance. However, most bagasse produced in Kenya is consumed internally for low-efficiency cogeneration in sugar mills, with surplus quantities remaining underutilized or disposed of (Food Business, 2025; Sharon Atieno, 2022).

Bagasse possesses a calorific value ranging between 7,500 and 9,000 kJ/kg when dried, enabling its application as a renewable energy source (Kabeyi & Olanrewaju, 2023). Globally, sugarcane bagasse has an estimated electricity generation potential of approximately 135,029 GWh per year (Ojo-kupoluyi et al., 2024). When extrapolated to Kenya's annual bagasse output, this corresponds to an estimated potential of roughly 650 GWh per year, equivalent to approximately 5% of the country's annual electricity demand (Tiema et al., 2024).

Beyond energy applications, bagasse supports multiple industrial value chains. The bagasse-based packaging market was valued at USD 210.2 million in 2024 and is projected to reach USD 361.9 million by 2030, reflecting a compound annual growth rate (CAGR) of 9.5%, with the Asia-Pacific region accounting for 45.8% of global demand (Grand View Research, 2024). In pulp and paper production, bagasse-based facilities in African contexts have demonstrated returns on investment of approximately 45% with payback periods of about 2.2 years (Manyuchi & Nkomo, 2015). At the smallholder level, bagasse-based mushroom cultivation in Kenya has been shown to achieve profitability within two production cycles with relatively modest capital input (Yakazi Network, 2024).

Despite this substantial economic potential, the effective utilization of bagasse is constrained by its high moisture content and susceptibility to microbial degradation. Without stabilization through appropriate treatment, bagasse rapidly deteriorates, reducing its suitability for high-value applications. Microbial instability, therefore, represents a critical bottleneck in transitioning bagasse from a low-value by-product to a reliable industrial feedstock. This constraint highlights the importance of controlled thermal treatment processes, which are examined in the subsequent sections of this chapter.

2.2 Industrial Treatment of Bagasse and Existing Limitations

In industrial bio-composite insulation production systems, bagasse undergoes controlled preparation before further processing. The material is typically soaked to achieve the required moisture content, drained to a wet but non-dripping condition, and subjected to thermal pasteurization at moderate temperatures, generally between 70 and 80 °C. The purpose of pasteurization in this context is not sterilization but selective microbial suppression. Competing microorganisms must be reduced while maintaining sufficient moisture to support subsequent biological binding processes.

However, current industrial practice often relies on empirical methods rather than thermally optimized designs. Pasteurization durations are frequently determined through experience rather than through heat transfer modeling. In some operations, steam exposure may extend up to 24 hours under the assumption that prolonged heating ensures microbial reduction. Such approaches introduce limitations, including low throughput, uncertain internal temperature distribution, and potential inefficiencies in steam utilization.

Furthermore, these systems typically do not incorporate measured thermophysical properties of bagasse under processing conditions. Parameters such as bulk density in the loose wet state, thermal conductivity at elevated moisture levels, and effective specific heat capacity are rarely determined experimentally for design purposes. The absence of property-based design frameworks limits predictive control and scalability of pasteurization systems.

2.3 Biomass Pasteurization and Technical Challenges

Pasteurization of lignocellulosic biomass involves heating to moderate temperatures, typically between 60 and 80 °C, for a sufficient duration to suppress undesirable microorganisms. Steam is widely used as the heating medium due to its high latent heat of vaporization, which enhances energy transfer during condensation (Hendriks & Zeeman, 2009).

In porous biomass beds such as sugarcane bagasse, heat transfer occurs through steam condensation at the solid surface, convective heat transfer within void spaces, and conduction through the solid matrix. The porous structure and low intrinsic thermal conductivity of bagasse present challenges for uniform heating. Reported thermal conductivity values for agricultural residues range between 0.05 and 0.30 W/m·K, depending on moisture content and structure (Tahir et al., 2021). Low conductivity restricts internal heat penetration, particularly in bulk beds.

Moisture content further influences thermal behavior. Fresh bagasse commonly exhibits moisture levels between 45–55% (wet basis) (Celignis Analytical, 2024; Lima, 2016). Because water has a higher specific heat capacity than dry biomass, increased moisture raises the effective heat capacity of the composite material. Consequently, more thermal energy is required to raise the temperature of wet bagasse to pasteurization conditions.

Bulk density also affects pasteurization performance. Loosely packed biomass exhibits high porosity and variable steam flow paths, potentially leading to non-uniform temperature profiles. Internal temperature gradients may develop if conductive resistance within the solid matrix is significant. While classical heat transfer theory provides tools such as transient energy balances and Biot number analysis to evaluate such systems, these approaches are seldom applied directly in practical biomass pasteurization operations.

Most published studies on biomass pretreatment emphasize biological performance metrics such as contamination reduction or yield enhancement (Koul et al., 2022; Schwerdtner Máñez et al., 2024), rather than thermodynamic modeling or steam behavior analysis. As a result, limited guidance exists for designing pasteurization systems based on heat transfer principles.

2.4 Literature Gaps and Pathways for Bridging the Gaps

The review of existing literature highlights several gaps relevant to engineering-based pasteurization system design. First, thermos-physical property data for sugarcane bagasse are often generalized for dry biomass or combustion applications, with limited state-specific characterization under wet processing conditions. Since pasteurization is conducted on soaked and drained bagasse, property values under these conditions are necessary for accurate modeling.

Second, bulk density measurements corresponding to loose industrial packing states are not consistently reported. Reactor sizing and heat transfer predictions depend strongly on these parameters.

Third, steam behavior within porous biomass beds is rarely integrated into transient heat transfer models. Although steam-based heating relies on latent heat release, many practical systems treat heating as uniform and do not consider steam quality evolution or condensation accumulation within the matrix.

Finally, there is limited development of pasteurization models tailored to biomass pasteurization processes in small- to medium-scale industrial applications. Bridging these gaps requires systematic experimental characterization of bagasse properties under pasteurization-relevant conditions and integration of these properties into transient energy balance models. Incorporating steam behavior into simulation frameworks would enable the prediction of temperature profiles and pasteurization time as functions of operating parameters.

Addressing these deficiencies would transition biomass pasteurization from empirical practice to model-based design, improving throughput, predictability, and process control in bio-composite production systems.

CHAPTER THREE: METHODOLOGY

3.0 Overview

This chapter outlines the systematic procedure adopted in the development and analysis of the pasteurizer model, structured into four integrated stages. The study began with the determination of thermal properties and moisture content of the material, where key parameters such as thermal conductivity, specific heat capacity, density, and initial moisture content were experimentally measured and analytically evaluated to provide accurate input data for subsequent heat and mass balance calculations. These properties formed the foundation for the development of the pasteurizer model, which was formulated using fundamental principles of heat transfer, thermodynamics, and energy conservation; appropriate assumptions were clearly defined to simplify the system without compromising physical realism, and governing equations describing heat transfer between steam and the material were derived to predict temperature variation, energy requirements, and pasteurization time. Particular attention was given to modeling steam behavior within the system, including condensation dynamics, heat transfer coefficients, and potential deposition effects, with sensitivity analysis conducted to assess the influence of varying steam flow rates on pasteurization time. Finally, the developed mathematical model was implemented using computational tools to simulate system performance under different operating conditions, generating temperature profiles and pasteurization durations which were analyzed to evaluate system effectiveness and support optimization of the pasteurization process.

3.1 Determination of Sugarcane Bagasse Thermal Properties

3.1.1 Thermal Conductivity Measurement

The thermal conductivity of the bagasse was determined using a steady-state experimental approach in which a custom rig was constructed to maintain a constant temperature gradient across the sample. As illustrated in Figure 3.1.1, the base of the rig was heated using a hot plate to maintain a constant high temperature, T_h, while the top surface was exposed to ambient air, allowing heat loss through natural convection. Temperature readings were recorded at three distinct axial locations along the sample, and the resulting temperature distribution was fitted to the one-dimensional heat transfer model given in Equation 3.1.1 (a). The value of the bagasse thermal conductivity, k_b, was obtained by minimizing the deviation between the experimentally measured temperatures and those predicted by the model.

The governing temperature relationship is expressed as:

$$T(x) = Th - \frac{hx}{kb}\ (Tc - Tamb)$$

1: Equation 3.1.1 (a): 1D-Heat Transfer

Where T_c represents the temperature at the top of the sample, T_{amb} is the ambient air temperature, and h is the convective heat transfer coefficient. The coefficient h was estimated using natural convection correlations based on the Nusselt number, defined as:

$$\mathrm{Nu} = \frac{hL}{ka}$$

with the empirical correlation for natural convection given by

$$Nu = 0.54(GrPr)^{0.25}$$

2: Equation 3.1.1 (b): Nusselt Number Correlation

Here, Gr denotes the Grashof number, Pr the Prandtl number, k_a the thermal conductivity of air, and L the characteristic length of the rig.

1: Figure 3.1.1: Thermal Conductivity Rig

3.1.2 Specific Heat Capacity Measurement

A sample of sugarcane bagasse was weighed and recorded. A brass ball of known mass and specific heat capacity was also weighed. To keep the bagasse submerged in hot water during heating, the bagasse and brass ball were tightly wrapped together using a polythene sheet and further secured with a sellotape. The masses and specific heat capacities of the polythene and sellotape were known. A beaker of water was heated until it reached a stable temperature. The bagasse assembly (bagasse, brass ball, polythene, and sellotape) was then fully immersed in the hot water. The system was left for a sufficient time to allow the entire assembly to reach thermal equilibrium with the water. The final temperature of the hot water was recorded as T-hot. Cold water was measured and poured into the calorimeter. The initial temperature of the cold water was recorded as T-cold. The heated bagasse assembly was quickly and carefully removed from the hot water and immediately placed into the cold water inside the calorimeter. All efforts were made to minimize heat loss during the transfer. The calorimeter was then covered, and the mixture was allowed to reach thermal equilibrium. The final stabilized temperature was recorded as T-final.

$$\mathrm{Cb} = \frac{m_{wC_{w(T_{final}-T_{cold})}} - \left(m_{brassC_{brass}} + m_{pC_p} + m_{sC_s}\right)\left(T_{hot} - T_{final}\right)}{m_{b(T_{hot}-T_{final})}}$$

3: Equation 3.1.2: Energy Conservation

2: Figure 3.1.2 Specific Heat Capacity Rig

3.1.3 Moisture Content Measurement

The moisture content of the bagasse was determined using the oven-drying technique. A sample with a known initial mass was placed in an oven maintained at 105 °C and dried for approximately seven hours. The mass of the sample was recorded at hourly intervals until it reached a constant value, indicating that drying was complete. The moisture content on a wet basis, MC_{wb}, was then calculated using the initial mass, m_{init}, and the final dry mass m_{fin}.

$$MC_{wb} = \frac{m\ init - m\ fin}{m\ fin}$$

4: Equation 3.1.3 Moisture Content Wet Basis

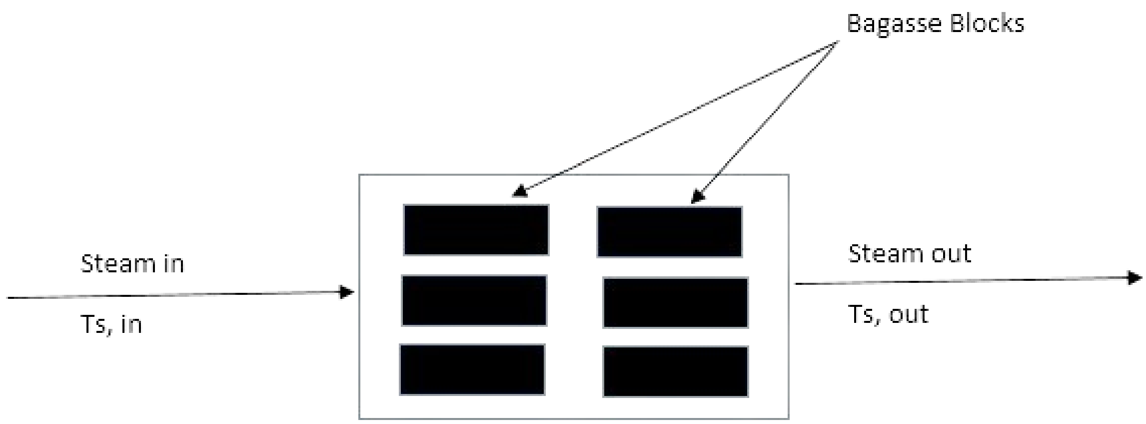
3.2 Development of the Pasteurization Model

3.2.1 Model Description

The pasteurization model was developed as a semi-batch process in which sugarcane bagasse was loaded once at the start, while steam was continuously supplied during heating. For simplicity, the bagasse was modeled as small, uniform blocks. A Biot number analysis showed that internal temperature gradients were negligible, allowing the use of the lumped capacitance assumption, where temperature varies only with time.

Direct contact between steam and bagasse was assumed, with heat transfer occurring mainly through steam condensation and convection. A transient energy balance was formulated to describe the temperature rise based on steam conditions, heat transfer coefficient, mass, and experimentally determined thermos-physical properties. Steam inlet and outlet conditions were included to account for energy exchange in the chamber.

Steam behavior and condensation effects were also incorporated into the model, enabling the prediction of temperature profiles and pasteurization time under different operating conditions.



3: Figure 3.2.1 Semi - Batch Operation

3.2.2 Model Assumptions

To simplify the mathematical formulation of the pasteurization system while maintaining physical realism, the following assumptions were adopted:

1. The system operates under semi-batch conditions, where bagasse is loaded once, and steam is supplied continuously during the heating period.
2. Direct contact occurs between steam and bagasse, allowing heat transfer primarily through condensation and convection at the solid surface.

3. The bagasse is modeled as small, uniform blocks to simplify geometry and enable representative heat transfer analysis.
4. A Biot number analysis was performed, and where $Bi<0.1$, the lumped capacitance approach was applied, assuming uniform temperature distribution within each bagasse block.
5. Thermo-physical properties (thermal conductivity, specific heat capacity, and density) are constant over the temperature range considered.
6. Steam temperature at the inlet is uniform and known.
7. The system pressure remains approximately constant during operation.

3.2.3 Biot Number Analysis

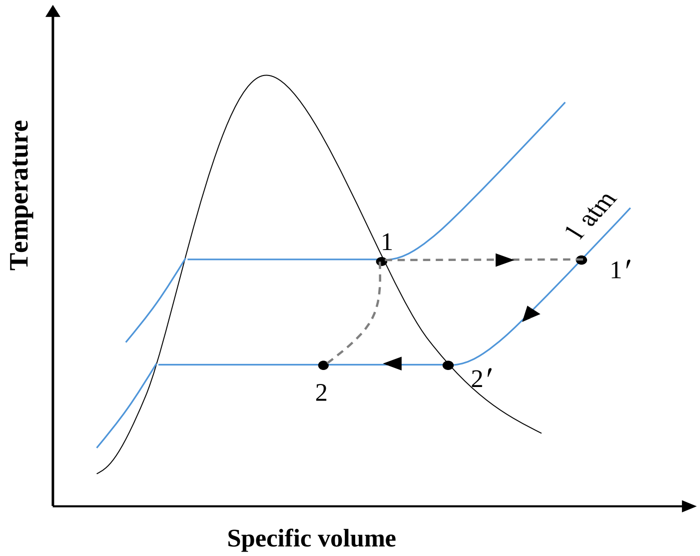
To justify the applicability of the lumped capacitance approach in modeling the transient heating of bagasse, a Biot number analysis was conducted to evaluate the relative magnitude of internal conduction resistance within the bagasse compared to the external convective resistance at the steam–solid interface. Steam was assumed to flow over rectangular bagasse blocks of length L_b, width w_b, and thickness t_b, with heat transfer occurring at the exposed surface. The Biot number, defined as the ratio of internal conductive resistance to external convective resistance, was evaluated using estimated heat transfer coefficients obtained from appropriate Nusselt correlations for forced convection. By establishing a geometrical criterion that ensures $Bi \leq 1$ (or $Bi \leq 0.1$ under stricter conditions), it was demonstrated that temperature gradients within the bagasse blocks are negligible relative to the surface temperature difference. This validates the assumption of spatially uniform temperature within each block, thereby permitting the use of the lumped capacitance method for transient analysis. A detailed derivation of the Biot number formulation and the resulting dimensional constraints is provided in Appendix A.

3.2.4 Governing Energy Balance

Referring to Figure 3.2.1, the governing equations for the pasteurization model were formulated by applying the fundamental principles of mass and energy conservation to both the steam and bagasse phases within the pasteurization chamber. Separate mass and energy balances were established for each phase, accounting for steady-steam flow, transient heating of the solid phase, and heat transfer due to sensible cooling and condensation effects. These balances were subsequently combined to obtain the dynamic equation describing the temperature evolution of the bagasse during pasteurization. See Appendix B for a detailed derivation of the mass and energy balance equations.

3.3 Steam Behavior Modeling

The thermodynamic behavior of steam between the inlet and outlet of the pasteurization chamber was analyzed to accurately describe the energy transferred to the bagasse. The steam process was divided into distinct stages based on its thermodynamic path, as shown in Figure 3.3.



4: Figure 3.3 T - v diagram for steam state transition

Upon entering the chamber, steam first undergoes an isenthalpic expansion process (Process 1–1′), during which pressure adjustment occurs without a change in specific enthalpy. This accounts for minor pressure variations between the supply line and the pasteurization chamber. Following expansion, the steam experiences sensible cooling (Process 1′–2′) as it transfers heat to the relatively cooler bagasse surface. In this stage, the temperature of steam remains above saturation temperature ($Ts > Tsat$) without phase change, and heat transfer occurs primarily through convection, and no accumulation of steam within the unit; hence, one can write:

$$\frac{dm\_s}{dt} = 0$$

Therefore,

$$\dot{m}\mathrm{s, in} = \dot{m}s, out$$

5: Equation 3.3 Steam mass flow rate

As cooling continues and the steam reaches saturation conditions at the operating pressure (approximately 1 atm), condensation begins. The condensation process (Process 2′–2) occurs at approximately constant temperature and represents the dominant heat transfer mechanism in the system due to the release of latent heat. This phase-change process significantly enhances the rate of energy transfer to the bagasse and governs the overall pasteurization time. Two condensation scenarios were considered in the model.

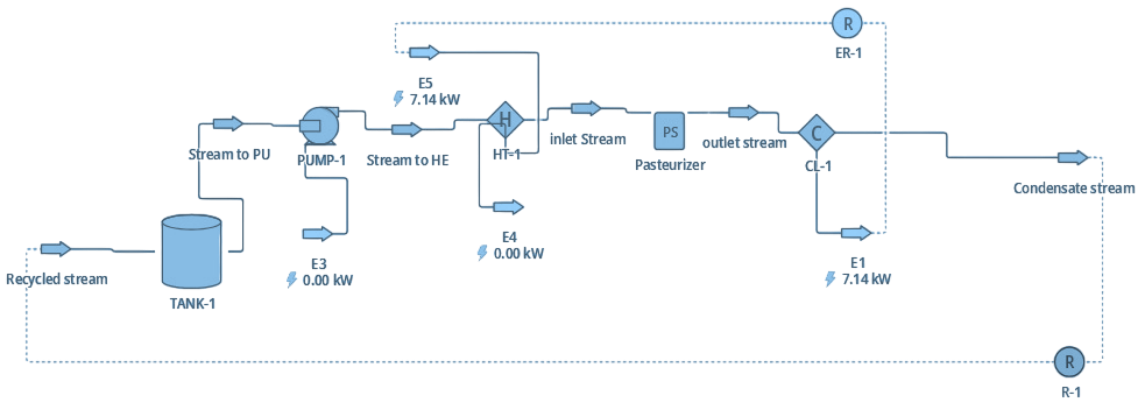
In the first scenario, condensation with deposition effects was assumed. Here, condensed steam accumulates within the bagasse structure, resulting in an increase in bagasse mass over time. Under this condition, steam exiting the chamber is assumed to be dry saturated vapor (quality equal to unity), as the condensed fraction is retained within the solid matrix. See appendix C.

In the second scenario, condensation without deposition effects was considered. In this case, no net increase in bagasse mass occurs, and the condensed liquid does not accumulate within the material. Consequently, the outlet stream consists of a two-phase steam–water mixture with variable quality depending on the extent of condensation. (See Appendix D)

These two scenarios were incorporated into the energy balance formulation by modifying the outlet steam enthalpy term, h_{out}, in the governing equation. The difference between inlet and outlet enthalpy accounts for both sensible and latent heat contributions. In particular, the latent heat component was quantified through the rate of condensation (See Appendix E), which represents the mass of steam undergoing phase change per unit time. Since the energy released during condensation is proportional to the product of the condensation rate and the latent heat of vaporization, incorporating this term enabled direct evaluation of how condensation behavior influences the heating rate of the bagasse and, consequently, the overall pasteurization time.

3.4 Model Implementation and Simulation

The developed mathematical model was implemented and simulated using DWSIM, an open-source chemical process simulation software, to evaluate the transient thermal behavior of the pasteurization system. Python script unit operation was used in modelling of the governing energy balance equations derived in Section 3.2, together with the steam behavior and condensation models described in Section 3.3, which formed the core of the simulation framework, see Figure 3.4. Experimentally determined thermos-physical properties of bagasse (thermal conductivity, specific heat capacity, and density) were incorporated as input parameters, while steam properties at saturation conditions were obtained within the DWSIM thermodynamic package.



6: Figure 3.4 Simulation Flowsheet

The pasteurization chamber was represented as a semi-batch thermal unit operation in which bagasse was treated as a solid phase with constant mass, while steam was modeled as a continuous flowing stream. The steam inlet conditions (temperature, pressure, and mass flow rate) were specified, and the outlet conditions were calculated based on energy exchange with the bagasse. The enthalpy difference between inlet and outlet steam streams was used to determine the rate of heat transfer to the bagasse, incorporating both sensible cooling and latent heat effects due to condensation. Depending on the selected condensation scenario (with or without deposition effects), the outlet steam quality and bagasse mass variation were adjusted accordingly within the simulation environment.

Transient temperature profiles of the bagasse were obtained by numerically solving the governing energy balance over time. Key performance indicators extracted from the simulation included pasteurization time, heat transfer rate, and the sensitivity of system performance to variations in steam flow rate. These results helped us foresee the thermal dynamics of the system and allowed the evaluation of optimal operating conditions.

The simulation outputs generated in DWSIM formed the basis of the analysis presented in Chapter Four, where the thermal performance of the pasteurization model, the influence of condensation behavior, and the effect of steam flow rate on pasteurization time and heating rate are discussed in detail.

CHAPTER FOUR: RESULTS, ANALYSIS, AND DISCUSSION

4.0 Overview

This chapter presents the experimental results obtained from the characterization of sugarcane bagasse, including thermal conductivity, specific heat capacity, moisture content, and bulk density. It then presents the simulation results from the developed semi-batch pasteurization model, followed by analysis and discussion of the findings.

4.1 Experimental Results

4.1.1 Thermal conductivity

Thermal conductivity was determined under steady-state conditions using a vertical conduction rig. The measured effective thermal conductivity of sugarcane bagasse is presented in **Table 4.1.1**

Thermal Conductivity (W/m.K)			
Dry Bagasse		**Wet Bagasse**	
Uncompressed	**Compressed**	**Uncompressed**	**Compressed**
0.07048	0.07664	0.2344	0.4168

1: Table 4.1.1 Thermal Conductivity Results

4.1.2 Specific heat capacity

Specific heat capacity was determined using calorimetric equilibrium methods. The measured values are presented in **Table 4.1.2**.

Specific Heat Capacity (J/Kg.K)	
Dry Bagasse	**Wet Bagasse**
1427.3	2510.1

2: Table 4.1.2 Specific Heat Capacity Results

4.1.3 Moisture content

Moisture content was determined through oven drying until constant mass was achieved. The corresponding wet mass, dry mass, and calculated moisture content are presented in **Table 4.1.3**.

Moisture Content (%)	
Dry Bagasse	**Wet Bagasse**
10	75

3: Table 4.1.3 Moisture Content Results

4.1.4 Bulk Density

Bulk density was determined by measuring the mass of bagasse occupying a known volume under typical packing conditions. The measured values are presented in **Table 4.1.4.**

Density (Kg/m^3)			
Dry Bagasse		**Wet Bagasse**	
Uncompressed	**Compressed**	**Uncompressed**	**Compressed**
90.4	130.4	330.6	537.4

4: Table 4.1.4 Bulk Density Results

4.2 Simulation Results from DWSIM

The semi-batch pasteurization model was simulated across a steam flow rate range of 0.1–200 kg/hr. For each steam flow rate, temperature–time profiles were computed, and pasteurization time was recorded.

The resulting pasteurization times for both condensation modelling scenarios are presented in **Table 4.2**.

Mass Flow rate (Kg/h)	Pasteurization Time(hrs)	
	Model 1 (No deposition)	Model 2 (Deposition)
0.1	32.90676	34.00113
0.5	25.75064	28.55705
2	16.00854	18.00266
4	10.30555556	12.219531
6	6.875	8.194444444
8	5.152777778	6.111111111
10	4.125	4.861111111
12	12.64253	4.166666667
14	13.52682	3.666666667
16	14.54006	3.333333333
18	15.70902	3.055555556
20	17.06287	2.916666667
50	25.62711	2.638888889
100	21.98724	2.5
150	18.8572	2.361111111
200	16.43115	2.361111111

5: Table 4.2 Simulation Results

4.3 Analysis

4.3.1 Steam Flow Rate Sensitivity Analysis

From the simulation results presented in **Table 4.2**, a sensitivity analysis was conducted to evaluate the variation in pasteurization time between the two modelling scenarios (condensation with deposition and condensation without deposition). The comparative differences in pasteurization times obtained from this analysis are summarized in **Table 4.3.1.**

Mass Flow rate (Kg/h)	Pasteurization Time(hrs)		
	Model 1 (No deposition)	Model 2 (Deposition)	Difference in pasteurization time
0.1	32.90676	34.00113	-1.09437
0.5	25.75064	28.55705	-2.80641
2	16.00854	18.00266	-1.99412
4	10.30555556	12.219531	-1.913975444
6	6.875	8.194444444	-1.319444444
8	5.152777778	6.111111111	-0.958333333
10	4.125	4.861111111	-0.736111111
12	12.64253	4.166666667	8.475863333
14	13.52682	3.666666667	9.860153333
16	14.54006	3.333333333	11.20672667
18	15.70902	3.055555556	12.65346444
20	17.06287	2.916666667	14.14620333
50	25.62711	2.638888889	22.98822111
100	21.98724	2.5	19.48724
150	18.8572	2.361111111	16.49608889
200	16.43115	2.361111111	14.07003889

6: Table 4.3.1 Sensitivity Analysis

Two distinct behavioral regions are observed:

Low Flowrate Region (0.1–10 kg/hr)

In this region, pasteurization time decreases gradually with increasing steam supply. The difference between the "with deposition" and "without deposition" model remain minimal. This indicates low sensitivity to condensation behavior.

High Flowrate Region (12–200 kg/hr)

Beyond 12 kg/hr, the divergence between the two modelling scenarios becomes increasingly pronounced. In the model with no deposition, the pasteurization times increase significantly, while in the model with deposition, the pasteurization times decrease gently. At the same time, the difference in predicted pasteurization time between the two models increases significantly.

This behavior indicates increasing sensitivity to condensation effects at higher steam flow rates.

4.3.2 Determination of Optimal Steam Flow Rate

The optimal steam flow rate was determined through sensitivity analysis by evaluating the variation of pasteurization time with steam flow rate under both condensation assumptions.

For each steam flow rate, the difference in pasteurization time between the deposition and non-deposition models was calculated. This difference served as a quantitative measure of sensitivity to condensation behavior.

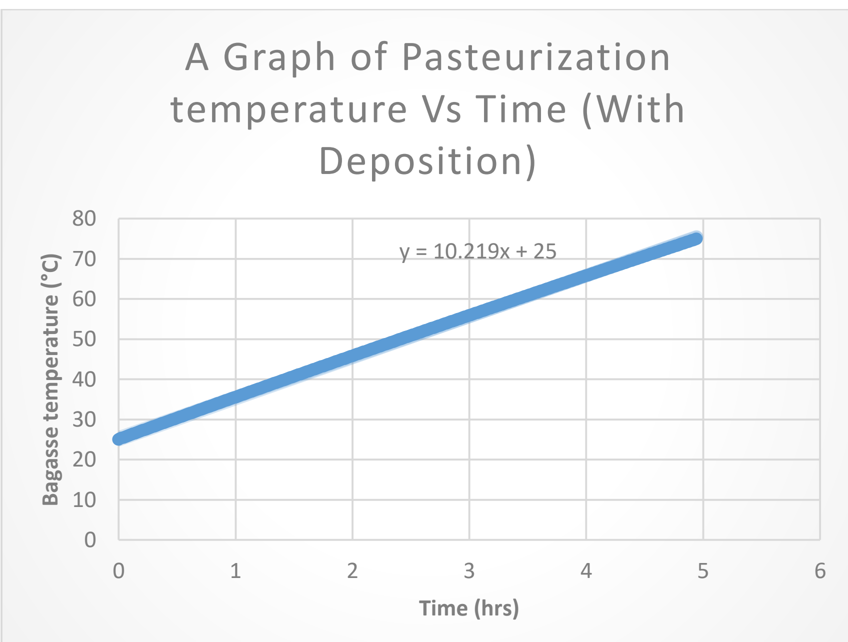
The selection of the optimal flowrate was based on two criteria:

1. Significant reduction in pasteurization time relative to steam flow rates.
2. Minimal divergence between condensation modelling assumptions.

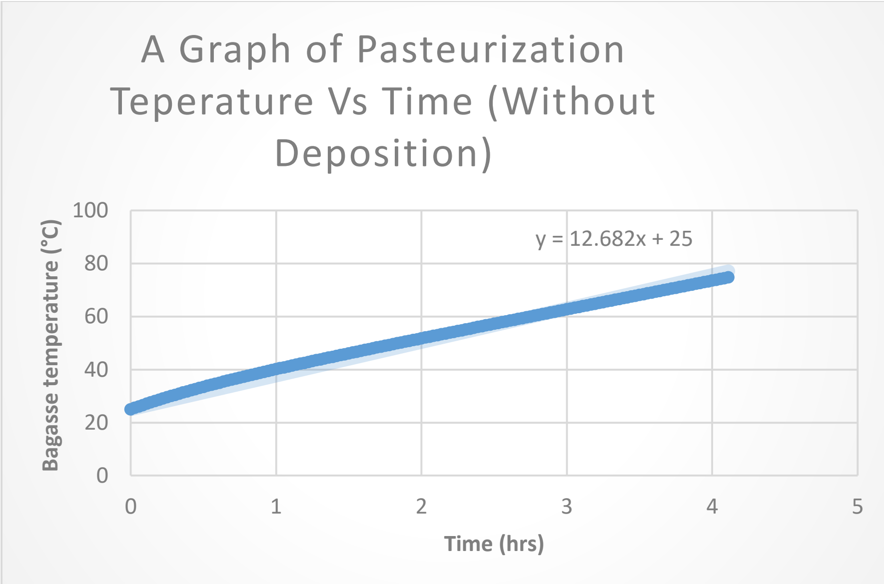
At **10 kg/hr**, pasteurization time was substantially reduced. The deposition model predicted 4.94 hours, while the non-deposition model predicted 4.11 hours. Although a difference exists, it is considerably smaller than the divergence observed at other steam flow rates.

Increasing the steam flow rate beyond 10 kg/hr resulted in a simultaneous increase in sensitivity to condensation effects. Therefore, 10 kg/hr was selected as the optimal operating condition based on the sensitivity analysis.

4.3.3 Temperature – Time Profile at Optimal Flow Rate



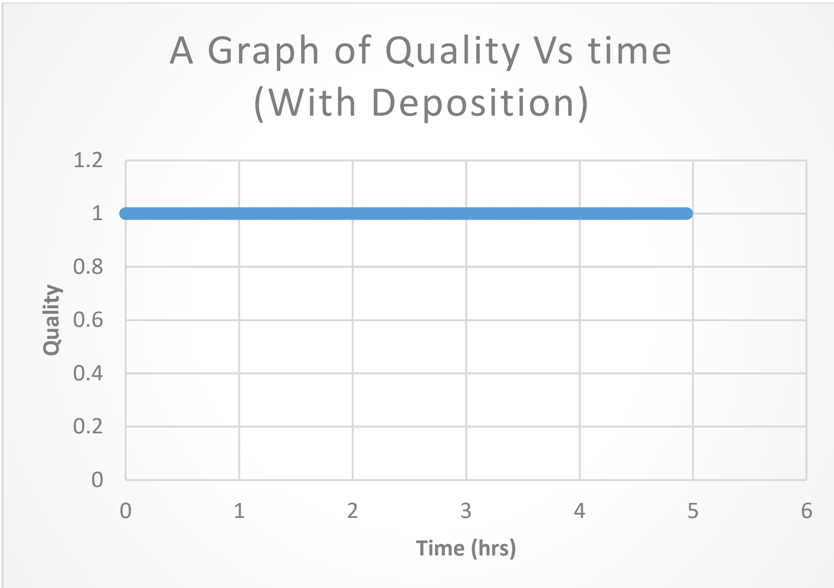
1: Chart 4.3.3(a) Temperature - Time Profile (Deposition)



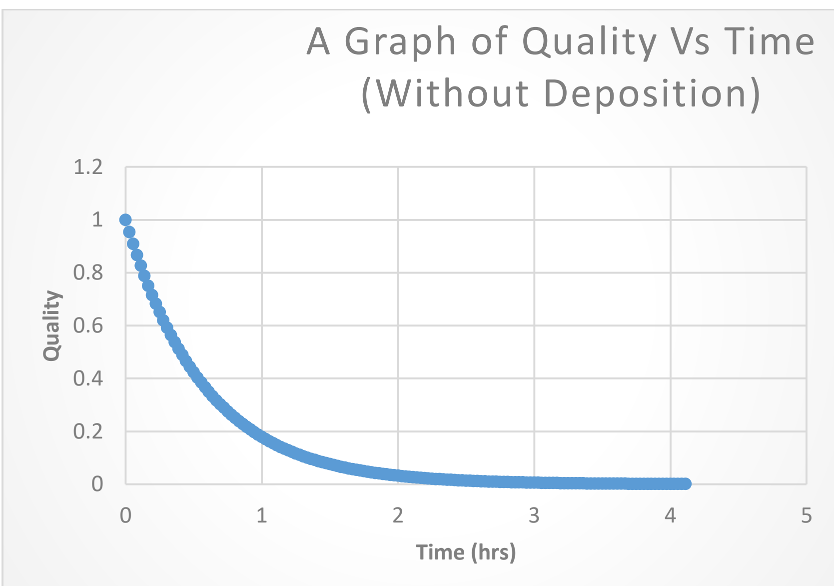
2: Chart 4.3.3(b) Temperature – Time Profile (No Deposition)

From the curves presented in charts 4.3.3 (a) and 4.3.3 (b), the curves portrayed are straight with positive gradients for both condensation models. This indicates that the rate of temperature increase remained constant throughout the heating period. The gradients of the curves represent the heating rates, where the model with deposition has a heating rate of 10.2 °C/hr, and 12.7 °C/hr for the one without deposition. The heating rates show the temperature rise in bagasse every hour.

4.3.4 Steam Quality Analysis



3: Chart 4.3.4 (a) Quality v Time (Deposition)



4: Chart 4.3.4 (b) Quality v Time (No Deposition)

From the curves presented in **Chart 4.3.4 (a)** and **4.3.4 (b)**, it is evident that steam quality differed between the two modelling scenarios. In the deposition model, condensed steam accumulated within the bagasse matrix, and the outlet steam remained as dry saturated vapor, resulting in a steam quality of unity throughout the process. In contrast, in the non-deposition model, the condensate is suspended in the exiting steam, leading to a steam quality value less than unity.

4.4 Discussion

The experimental characterization confirmed that sugarcane bagasse exhibits relatively low thermal conductivity, which is consistent with fibrous, porous biomass materials. The low conductivity is attributed to the high void fraction and the presence of trapped air within the matrix. Air acts as a thermal insulator, and the discontinuous solid–solid contact pathways within the fibrous structure further limit conductive heat transfer.

It was observed that wet bagasse exhibited higher thermal conductivity compared to dry bagasse. This behavior is explained by the displacement of air within pore spaces by water, which possesses significantly higher thermal conductivity than air. Consequently, moisture content contributed positively to heat conduction within the packed bed. Similarly, the results indicated that wet bagasse had a higher specific heat capacity than dry bagasse. This was expected because water has a relatively high specific heat capacity compared to dry fibrous biomass. Therefore, increasing the moisture content increased the overall energy required to raise the temperature of the bagasse.

Moisture content was incorporated into the mathematical model because it directly influences both thermal conductivity and specific heat capacity. Higher moisture levels improved conduction characteristics but simultaneously increased the heat storage capacity of the material. As a result, although wet bagasse transfers heat more effectively, it requires more energy to reach pasteurization temperature due to the additional thermal mass. This finding was particularly significant for the semi-batch model, since specific heat capacity directly determines the rate of temperature rise under steam heating.

The measured bulk density values indicated that uncompacted bagasse possessed relatively low density due to its high porosity. Bulk density is a critical design parameter in pasteurization system development because it determines the total mass of bagasse per unit volume.

The simulation results further reinforced the importance of these parameters. The temperature–time curves obtained from the model were straight lines with positive gradients for both condensation scenarios. This indicates that the rate of temperature increase remained approximately constant throughout the heating period. The linear behavior confirms that heat release from steam was steady over the heating period.

The optimal flow rate of 10 kg/hr was identified at the transition between these regions, where sufficient latent heat was supplied without excessive steam input. This confirms that system performance is governed primarily by heat transfer characteristics and phase-change dynamics rather than by excessive steam supply. At the optimal steam flow rate of 10 kg/hr, the heating rate without deposition (12.7 °C/hr) was higher than that with deposition (10.2 °C/hr), resulting in shorter pasteurization time (4.11 hours compared to 4.94 hours). The non-deposition case indicates improved heat transfer due to the absence of condensate accumulation.

Overall, the integration of experimentally determined thermal properties, moisture content, and bulk density into the semi-batch model significantly enhanced its predictive reliability. The results demonstrate that both material properties and steam thermodynamics must be carefully considered in pasteurizer design. The developed model, therefore, provides a scientific basis for designing a controlled semi-batch pasteurization system with predictable and optimized performance.

CHAPTER FIVE: PASTEURIZER SYSTEM DESIGN PROPOSAL, CONCLUSIONS, AND RECOMMENDATIONS

5.1 Pasteurizer System Design Proposal

The pasteurizer system design was proposed directly from the semi-batch pasteurization model and the experimentally determined thermal properties of sugarcane bagasse. The design parameters were selected based on simulation results, sensitivity analysis, and steam thermodynamic behavior.

5.1.1 Design Parameters

Parameter	Value	Units
Design steam flow rate	10	kg/hr
Operating pressure	2	bar
Steam temperature	121.212	°C
Bagasse moisture	75	%
Pasteurization temperature	75	°C
Pasteurization time	5	hrs
Bagasse throughput per batch	1000	kg
Chamber volume	3	m^3
Number of blocks	20	blocks
Boiler capacity	10	kg/hr

5: Chart 5.1.1 Design Parameters

5.1.2 Design Justification

The steam flow rate of 10 kg/hr was selected as the optimal value from sensitivity analysis, where it provided sufficient latent heat for heating without unnecessary steam supply. An operating pressure of 2 bar produces steam at approximately 121.2 °C, creating a strong temperature driving force for condensation and steady heat transfer. Although the target pasteurization temperature is 75 °C, the higher steam temperature ensures consistent heating.

The pasteurization time of 5 hours is based on model predictions showing linear temperature rise under constant steam supply.

A batch capacity of 1000 kg was selected to improve throughput compared to the previous 500 kg oven system. Based on measured bulk density, a chamber volume of 3 m^3 was determined to allow proper steam penetration while maintaining appropriate packing conditions.

The boiler capacity was matched directly to the design steam flow rate to ensure controlled and efficient operation.

5.1.3 Operating Principle

The proposed system operates under semi-batch conditions. Wet bagasse (75% moisture) is loaded into the insulated chamber at the beginning of the process. Saturated steam is supplied continuously at 10 kg/hr.

Steam undergoes sensible cooling and then condenses upon contact with the cooler bagasse. The latent heat released during condensation drives a steady, linear increase in temperature. Pasteurization is achieved after the bagasse reaches 75 °C, which is approximately 5 hours.

5.2 Conclusions

This study successfully developed a semi-batch pasteurization model for sugarcane bagasse and used it to guide the design of an improved pasteurizer system.

Experimental results showed that sugarcane bagasse has low thermal conductivity due to its porous structure, and that moisture content increases both thermal conductivity and specific heat capacity. Bulk density was identified as an important parameter because it is useful when modelling bagasse throughput.

The simulation demonstrated that heating is mainly driven by latent heat released during steam condensation. The temperature–time curves were straight lines with positive gradients, indicating steady heating under constant steam supply. An optimal steam flow rate of 10 kg/hr was identified for steady operation.

Compared to the previous boiler–oven system, the proposed design reduces pasteurization time from about 12 hours to 5 hours and increases batch capacity from 500 kg to 1000 kg. Steam supply is controlled and matched to demand, improving energy use and process consistency.

5.3 Recommendations

Further work is recommended to strengthen and expand the application of the developed pasteurization model.

A pilot-scale pasteurizer should be constructed to validate the simulation results under actual operating conditions. Experimental verification will confirm heating rates, pasteurization time, and steam consumption.

Additional research should investigate condensate management to determine whether active drainage or controlled retention provides better heating performance.

The effect of different packing densities and loading configurations should also be studied to better understand steam distribution and heating uniformity.

Finally, scale-up studies and economic evaluation should be conducted to assess industrial feasibility, including energy consumption and equipment sizing.

These next steps will support practical implementation and further improvement of the proposed pasteurizer system.

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APPENDICES

Appendix A: Biot Number Analysis

Heat transfer occurs at the exposed surface of the block due to forced convection and steam condensation. However, for the system to be treated as a lumped system (i.e., negligible internal temperature gradients within the solid), the Biot number must satisfy:

$$Bi = \frac{htb}{kb} \leq 1$$

5: Equation A.1

where k_b is the thermal conductivity of bagasse and h is the convective heat transfer coefficient between steam and the bagasse surface. The heat transfer coefficient h was estimated using a Nusselt number correlation for laminar forced convection over a horizontal flat plate:

$$Nu = \frac{hLb}{ks}$$

6: Equation A.2

With,

$$Nu = 0.664Re^{0.5}Pr^{0.333}$$

7: Equation A.3

where R_e is the Reynolds number, P_r is the Prandtl number, and k_s is the thermal conductivity of steam. Substituting the Nusselt relation into the Biot number expression gives:

$$Bi = 0.664Re^{0.5}Pr^{0.333}\left(\frac{ks}{kb}\right)\left(\frac{tb}{Lb}\right) \leq 1$$

8: Equation A.4

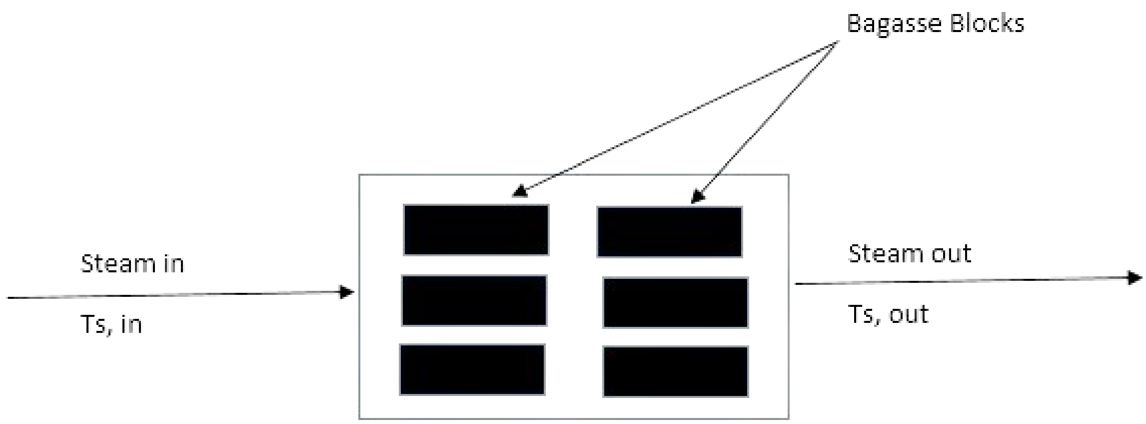
Using the upper Reynolds limit for laminar flow ($Re = 5\times10^5$), and assuming saturated steam at 1 atm with $P_r = 0.97$ and $k_s = 0.025$ W/mK, together with the experimentally measured thermal conductivity of wet uncompacted bagasse $k_b = 0.244$ W/mK, the geometrical constraint required to satisfy the lumped system assumption becomes:

$$\frac{tb}{Lb} \leq 0.01$$

9: Equation A.5

Achieving this criterion, bagasse blocks with $L_b = 1.485$m, $w_b = 0.965$ m and $t_b = 0.12$ m. Under these dimensions, each block would have a volume of 0.172 m^3 and, using an experimentally determined density of uncompressed wet bagasse of 330 kg/m^3, a mass of approximately 56.75 kg per block. For a processing capacity of 1 ton (1000 kg), approximately 20 such blocks would be required. The total heat transfer area available (assuming heat transfer occurs through the top surface only) would therefore be approximately 28 m², while the total occupied volume would be about 3.44 m³, which represents the minimum required volume of the pasteurization chamber, and this analysis confirms that, provided the geometrical criterion is satisfied, the lumped capacitance model is valid for the pasteurization system and can be used to reliably predict transient heating behavior.

Appendix B: Governing Equations



7: Figure B.1 Semi - Batch Model Illustration

Given Figure B.1, the governing equations for the pasteurization model were developed by applying fundamental principles of mass and energy conservation to both steam and bagasse phases within the pasteurization chamber.

A mass balance on the steam phase gives:

$$\frac{dms}{dt} = \dot{m}s,in - \dot{m}s,out$$

10: Equation B.1

where m_s is the mass of steam within the chamber and $\dot{m}$ represents the steam mass flow rate at the inlet and outlet. Since the system operates under steady-steam flow conditions (after initial transients), there is no accumulation of steam within the unit, and thus:

$$\frac{dms}{dt} = 0$$

Which implies:

$$\dot{m}s,\text{in} = \dot{m}s,out = \dot{m}_s$$

Similarly, for the bagasse phase, a mass balance yields:

$$\frac{dmb}{dt} = 0$$

Since the bagasse is loaded at the beginning of the process and remains inside the chamber throughout pasteurization (semi-batch operation). Thus, the mass of bagasse remains constant during heating. Energy balances were then formulated separately for steam and bagasse. For the steam phase:

$$m_sC_{v,s}\ \frac{dTs}{dt} = \dot{Q}_{out} + \dot{m}s\ (h_{in} - h_{out})$$

11: Equation B.2

And for the bagasse phase:

$$m_bC_{p,b}\ \frac{dTb}{dt} = \dot{Q}_{in}$$

12: Equation B.3

where $c_{v,s}$ is the specific heat capacity of steam at constant volume, $c_{p,b}$ is the specific heat capacity of bagasse at constant pressure, T_s and T_b are the temperatures of steam and bagasse respectively, and h_{in} and h_{out} are the specific enthalpies of steam at inlet and outlet. The heat transfer terms satisfy:

$$\dot{Q}_{\text{out}} = - \dot{Q}_{\text{in}}$$

Since heat lost by steam is gained by bagasse.

Under steady-state steam operation (i.e., steam temperature variations within the chamber are negligible after startup), the accumulation term in the steam energy balance becomes zero:

$$m_sC_{v,s}\ \frac{dTs}{dt} = 0$$

Consequently, combining the steam and bagasse energy balances yields the governing equation of the pasteurization model:

$$m_bC_{p,b}\ \frac{dTb}{dt} = \dot{m}s\ (h_{in} - h_{out})$$

13: Equation B.4

This equation states that the rate of energy accumulation within the bagasse equals the rate of enthalpy transfer from the flowing steam. It forms the fundamental dynamic relationship used to predict the transient temperature rise of bagasse during pasteurization. The term $\dot{m}s\ (h_{in} - h_{out})$ represents the net thermal energy delivered by steam, including both sensible and latent heat contributions due to condensation, and this governing equation served as the foundation for subsequent numerical simulation and determination of pasteurization time under varying steam flow rates.

Appendix C: Modelling condensation with deposition

Here, condensed steam was assumed to accumulates within the bagasse structure, resulting in an increase in bagasse mass over time. Under this condition, steam exiting the chamber is assumed to be dry saturated vapor (quality equal to unity), as the condensed fraction is retained within the solid matrix, and thus a mass balance for steam was written as:

$$\frac{dms}{dt} = \dot{m}s, in - \dot{m}s, out - R_{cond}$$

14: Equation C.1

Where, R_{cond} is the rate of condensation, however, since there's no accumulation of steam within the unit then, the first term goes to zero, and we wrote the mass balance as:

$$\dot{m}s, out = \dot{m}s, in - R_{cond}$$

15: Equation C.2

Then, since there's a change in the mass of bagasse, the mass balance equation was written as:

$$\frac{dmb}{dt} = R_{cond}$$

16: Equation C.3

And from this equation, it can clearly be seen that the effective mass of bagasse increases with the rate of steam deposition.

The energy balances for both steam and bagasse putting into consideration that steam undergoes constant-temperature cooling was written as:

$$0 = \dot{Q}_{out} + \dot{m}_{s,in}h_{in} - \dot{m}_{s,out}h_{out}$$

17: Equation C.4

$$\frac{d(mpCp,bTb)}{dt} = \dot{Q}_{in}$$

18: Equation C.5

Combining equation C.5 and equation C.2, then we obtain:

$$\dot{Q}_{out} = -(\dot{m}_{s,in}[h_{in} - h_{out}] + R_{cond}h_{out})$$

19: Equation C.6

Equation C.5, was seen to involve two rate terms, one

for the rate of change of energy of bagasse due to increasing temperature and

the other for rate of change of energy of bagasse due to increasing mass.

Combining equation C.5 with equation C.6, this gives the final form of the energy equation for bagasse as (while putting into consideration that $\dot{Q}_{out} = -\dot{Q}_{in}$):

$$mpCp, b\frac{dTb}{dt} = \dot{m}_{s,in}(h_{in} - h_{out}) + R_{cond}(h_{out} - c_{p,b}T_b)$$

20: Equation C.7

Appendix D: Modelling condensation without deposition

In this case, no net increase in bagasse mass occurs, and the condensed liquid does not accumulate within the material but it is assumed to remain within the steam. Consequently, the outlet stream consists of a two-phase steam–water mixture with variable quality (quality, $\bar{x} \leq 1$) depending on the extent of condensation, and thus mass balances for steam by considering the vapor portion, and liquid portion was written as:

$$\frac{d(msx)}{dt} = \dot{m}_{s,in}x_{in} - \dot{m}_{s,out}x_{out} + R_{cond}$$

21: Equation D.1

$$\frac{d(ms[1-x])}{dt} = 0 - \dot{m}_{s,out}(1 - x_{out}) + R_{cond}$$

22: Equation D.2

The quality of steam at inlet was assumed to be one (x_{in} = 1) since saturated vapour is produced by the boiler. In addition, it was assumed that there is good mixing within the unit such that $x_{out} = x = \bar{x}$. That is, a single value, $\bar{x}$, can be used to represent the quality of steam within the unit and at the outlet, and also by assuming that there was no accumulation of steam within the unit, then equation D.1 and D.2 above was combined to give the following equation for the quality of steam:

$$m_s\frac{dx}{dt} = \dot{m}_s(1-\bar{x}) - R_{cond}$$

23: Equation D.3

And for bagasse, the mass balance equation was written as:

$$\frac{dmb}{dt} = 0$$

24: Equation D.4

And from this equation, it can be seen that the mass of bagasse remains constant. The energy balances for steam and bagasse was written as:

$$\frac{d(ms[uf+ufg])}{dt} = \dot{Q}_{out} + \dot{m_s}(h_{in} - h_{out})$$

25: Equation D.5

$$m_{pCp.b}\frac{dTb}{dt} = \dot{Q}_{in}$$

26: Equation D.6

The rate term of equation D.5 is expressed in terms of the specific internal energy of the liquid portion of steam (uf), the specific internal energy of the vapor portion of steam (ug) and their difference ($u$fg = ug − uf). Since there is no accumulation of steam within the unit and the temperature of steam is constant (hence specific internal energies are constant), the rate term was expressed in terms of the rate of change of the quality as shown in equation D.5. In addition, equation D.5 was combined with equation D.3 to give the following final expression for $\dot{Q}_{out}$.

$$\dot{Q}_{out} = -(\dot{m}_{s,in}[h_{in} - h_{out}] + [R_{cond} - \dot{m_s}(1 - \overline{x})]ufg)$$

27: Equation D.7

Now since $\dot{Q}_{out} = -\dot{Q}_{in}$, then equation D.7, was introduced to equation D.6, giving the final energy expression for bagasse as:

$$m_{pCp.b}\frac{dTb}{dt} = (\dot{m}_{s,in}[h_{in} - h_{out}] + [R_{cond} - \dot{m_s}(1 - -\overline{x})]ufg)$$

28: Equation D.8

Appendix E: Modelling the Rate of Condensation

The condensation rate was modeled by assuming film condensation occurring within a closed and thermally isolated system. Under these conditions, applying an energy balance leads to the following expression:

$$Rcond = \frac{hA(Tsat - Tb)}{hfg}$$

29: Equation E.1

where h_{fg} denotes the latent heat of vaporization, and h is the condensation heat transfer coefficient. The heat transfer coefficient was estimated using the Nusselt theory for laminar film condensation on a horizontal surface, given by:

$$h = 0.729[\frac{(g\rho L^\wedge 2h\_fgkL^\wedge 3)}{\mu L(Tsat - Tb)Lb}]^\wedge 0.25$$

30: Equation E.2

where ρ_L, k_L and μ_L represented the density, thermal conductivity and viscosity of the liquid water respectively while g represented the gravitation constant.